

# Fair Clustering

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*Fairness-Aware Clustering with KL Divergence & Continuous Sensitive Attributes*

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# Why Does Fairness in Clustering Matter?



### Algorithmic Bias

Standard k-means optimises cluster compactness only — ignoring demographic distributions. Minority age groups get underrepresented clusters.



### Real-World Impact

Biased clusters affect credit scoring, job recommendations, healthcare triage, and targeted policy — decisions that shape lives.



### Research Gap

Prior work treats sensitive attributes as categorical (race, gender). Continuous attributes like age require a different probabilistic framework.

**Objective: Cluster data so that each cluster's sensitive attribute distribution mirrors the global distribution.**



## Problem Statement

### Formal Problem Definition

#### Given:

- Dataset  $X = \{x_1, \dots, x_n\} \in \mathbb{R}^d$
- Sensitive attribute  $s = \{\text{age}\} \in \mathbb{R}$  (continuous)
- Number of clusters  $k$
- Fairness weight  $\lambda \geq 0$

#### Minimise:

$$L(C) = \text{Utility}(C) + \lambda \cdot \text{Fairness}(C)$$

*Subject to:  $C_j$  partition  $X$  into  $k$  clusters*

#### Components:

##### Utility = SSE

Sum of squared distances from each point to its cluster centroid.  
Standard k-means objective.

##### Fairness = $\sum \text{KL}(P_{\text{global}} \parallel P_{\text{cluster}})$

KL divergence between global age distribution and per-cluster age distribution.

##### $\lambda$ (Lambda)

Controls the fairness–utility tradeoff.  $\lambda=0$  gives standard k-means; higher  $\lambda$  enforces fairness.



# Variational Fair Clustering — Key Theoretical Framework

### *Prior Art & Limitations*

#### **Balanced Clustering (Fairlets., 2017)**

First fairness-aware clustering. Categorical sensitive attributes only. Uses fair representation constraints.

#### **Socially Fair k-Means (Abbasi et al., 2021)**

Minimises maximum group cost. Still limited to discrete groups.

#### **Scalable Fair Clustering (Backurs et al., 2019)**

Polynomial-time approximation for categorical fairness. Does not extend to continuous attributes.



## Literature Review

### *Theoretical Tools Used*

#### Auxiliary Function Approach

Constructs a surrogate objective  $Q(\theta, \theta^{\text{old}})$  that upper-bounds the true loss. Optimising  $Q$  guarantees monotone decrease of the original objective — similar to EM algorithm convergence.

#### Auxiliary Function $A_i(S)$

$$A_i(S) = \sum_p s_p^T (a_{ip} + \lambda b_{ip} + \log s_p - \log s_{ip})$$

$a_{ip}$  = K-means term:  $\|x_p - c_{ik}\|^2$

$b_{ip}$  = fairness term (Lipschitz gradient)

$\log s_p$  = entropy barrier (handles simplex)

**Closed-form update (KKT):**

$$s_i^{+1} = s_{ip} \exp(-(a_{ip} + \lambda b_{ip})) / \text{normalizer}$$



## Literature Review

### Theoretical Tools Used

#### Concave–Convex Procedure (CCCP)

Decomposes the objective into convex + concave parts.  
Linearises the concave part at each iteration, creating a sequence of convex sub-problems guaranteed to converge.

#### Variational Bound Tightening

At each step, the auxiliary Q is made tight at the current solution ( $Q = L$ ). This ensures the update strictly improves or maintains the objective value.

#### MM Convergence Guarantee

$E(S_i^{+1}) \leq A_i(S_i^{+1}) \leq A_i(S_i) = E(S_i)$   
→ Objective never increases each iteration.  
→ Converges to local minimum.

#### Clustering Objective Potentials

| Method    | $a_{ip,k}$                                             |
|-----------|--------------------------------------------------------|
| K-means   | $  x_p - c_{ik}  ^2$                                   |
| K-medians | $d(x_p, c_{ik})$                                       |
| Ncut      | $dpz_{ik} - 2\sum w(x_p, x_q) s_{i,q,k} / d^T S_{i,k}$ |



Our Contributions vs Prior Work

| Aspect                 | Prior Work                 | Our Work                                   |
|------------------------|----------------------------|--------------------------------------------|
| Sensitive Attribute    | Categorical (race, gender) | Continuous (age)                           |
| Fairness Metric        | Balance ratio / max-cost   | KL Divergence                              |
| Optimal k Selection    | Manual / heuristic         | Silhouette score, elbow method auto-select |
| Initialisation         | Random / k-means           | K-Means++ for stability                    |
| $\lambda$ Tuning       | Fixed or not explored      | Sweep $\lambda \in [0, 6]$                 |
| Attribute Distribution | Discrete histograms        | Continuous KDE                             |



# Fair K-Means Algorithm

1

### Initialise:

Run K-Means++ to select  $k$  centroids. Compute global age distribution  $P_{\text{global}}$ .

2

### Assign Points:

For each point  $x_i$ , try all  $k$  clusters.  
Score =  $\text{dist}(x_i, \mu_j) + \lambda \cdot \text{KL}(P_j \parallel P_{\text{global}})$ . Assign to argmin cluster.

3

### Update

#### Centroids:

Recompute each centroid as the mean of assigned points. Skip empty clusters (rare with K-Means++ init).

4

### Check

#### Convergence:

If cluster assignments unchanged from previous iteration, stop.  
Otherwise go to Step 2.

5

### Evaluate:

Compute Utility (SSE) and Fairness (total KL divergence). Store results for current  $\lambda$ .

### Pseudocode

```
Input:  $X, s, k, \lambda, \text{max\_iter}$   
 $\mu \leftarrow \text{KMeans++}(X, k)$   
 $c \leftarrow \text{argmin}_j \text{dist}(X, \mu_j)$   
  
for iter in 1..max_iter:  
     $\mu_j \leftarrow \text{mean}(X[c==j]) \forall j$   
    for each  $x_i$ :  
        for j in 0..k:  
             $d = \|x_i - \mu_j\|$   
             $f = \text{KL}(P_j \parallel P_g)$   
            cost =  $d + \lambda \cdot f$   
             $c[i] = \text{argmin cost}$   
        if c unchanged: break  
  
return c,  $\mu$ , utility, fairness
```



## Dataset & Cluster Selection

### UCI Adult Dataset | Selecting Optimal k

**~45,000**

Instances

**14**

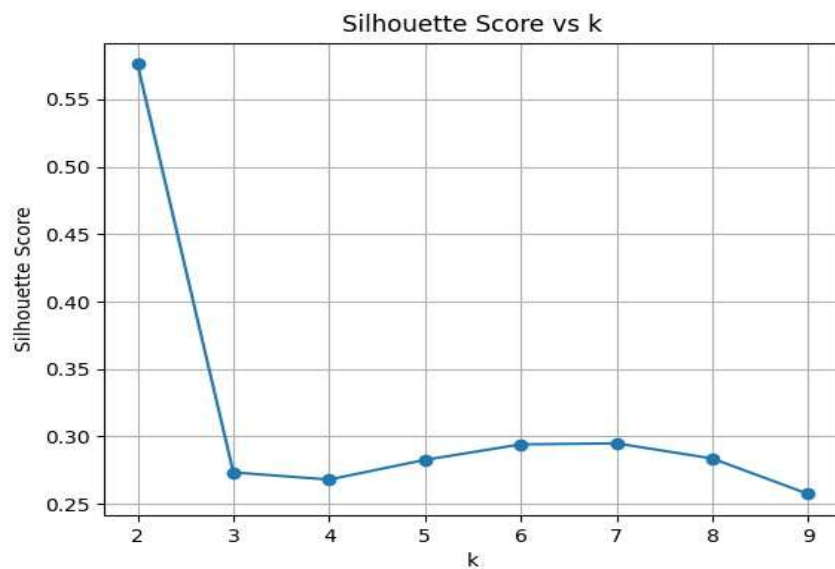
Features

**Age**

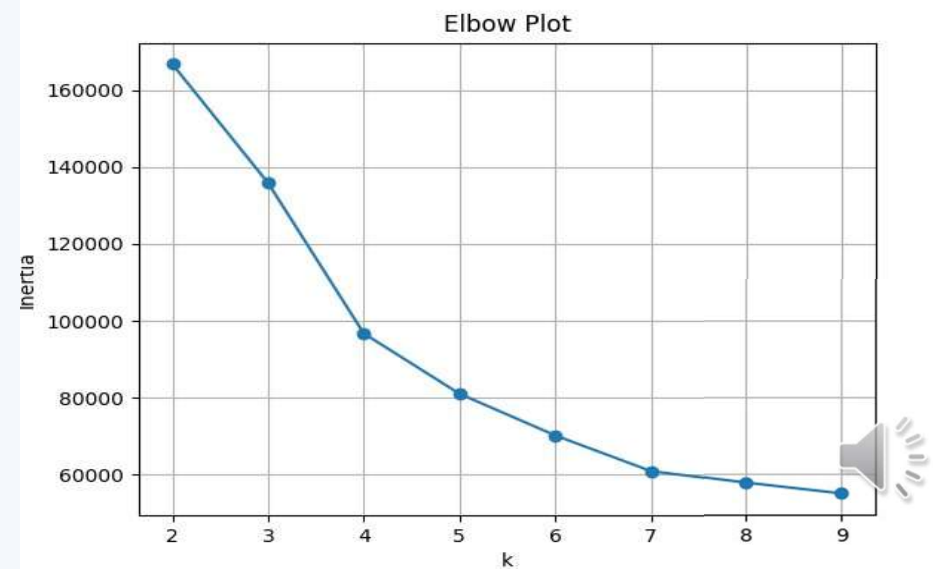
Sensitive  
Attribute

**4**

Best k



*Silhouette Method (left): k=2.*



*Elbow Method (right): confirms k=4.*

### How Points Are Assigned in Fair K-Means

#### Assignment Score Function

$$\text{score}(i, j) = \text{dist}(x_i, \mu_j) + \lambda \cdot \text{fairness}(j)$$

where:

$$\text{dist}(x_i, \mu_j) = \|x_i - \mu_j\|_2 \text{ (Euclidean)}$$

$$\text{fairness}(j) = \text{KL}(P_{\text{cluster}_j} \parallel P_{\text{global}})$$

$$\text{KL}(P \parallel Q) = \sum_a (P(a) + \epsilon) \cdot \log[(P(a) + \epsilon) / (Q(a) + \epsilon)]$$

$P_{\text{global}}$ : age of full dataset

$P_{\text{cluster}_j}$ : age of cluster  $j$

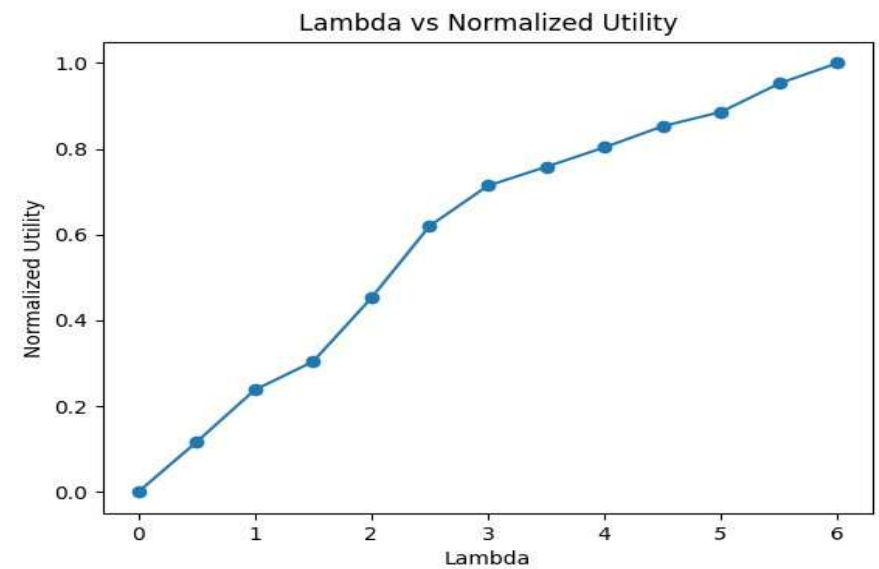
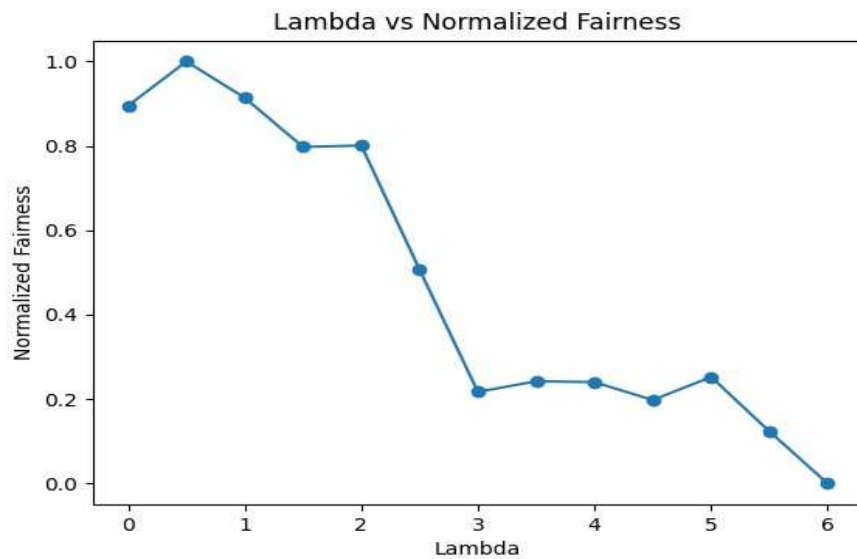
#### What This Achieves

1. Every point is tried in each cluster — greedy per-point assignment.
2. A cluster whose age distribution deviates from global incurs a higher fairness penalty, discouraging biased grouping.
3.  $\lambda=0$  recovers standard k-means (no fairness constraint).
4. Higher  $\lambda$  forces clusters to mirror the global age distribution, at the cost of higher SSE.



## Results — Lambda Tradeoff

### $\lambda$ Sweep: Utility vs Fairness Tradeoff



Normalized KL Fairness (left): Decreases as  $\lambda$  increases — clusters become more age-balanced.

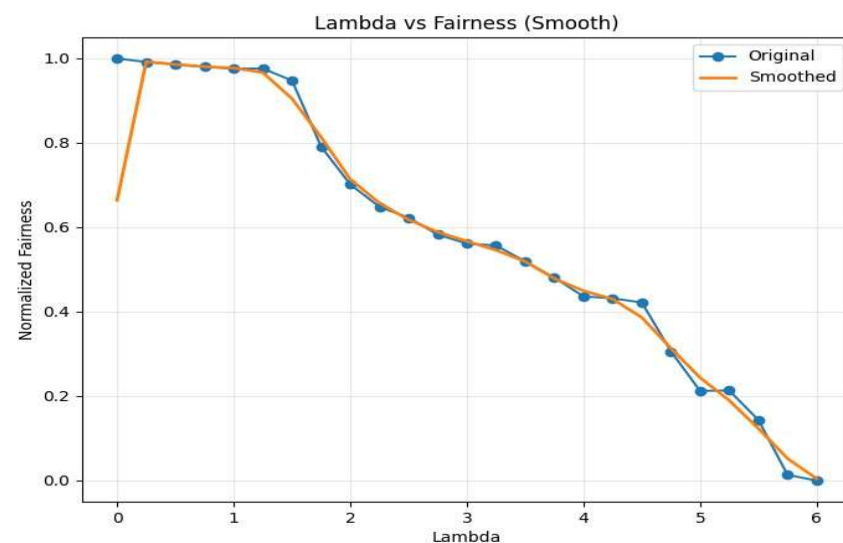
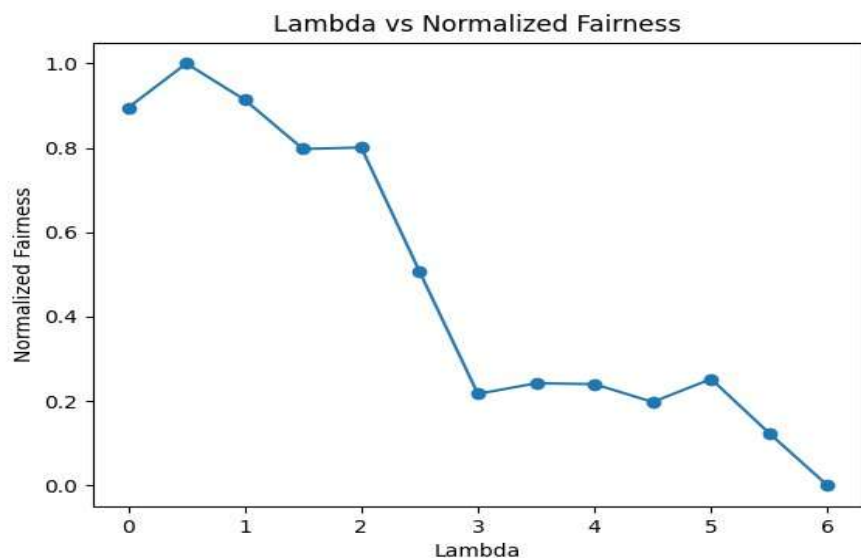
Normalized Utility (right): Increases monotonically — higher  $\lambda$  forces fewer compact clusters (utility sacrifice).



| Avg Utility = 176976.12 | Avg Fairness (KL) = 0.0883 | Score = min(utility\_norm + fairness\_norm)

## Results — Tradeoff Analysis

### Normalized Fairness & Smooth Tradeoff Curves



Normalized Fairness (left): Non-monotone initially — small  $\lambda$  can temporarily increase KL as cluster centroids shift. Drops sharply after  $\lambda \approx 2.5$ .

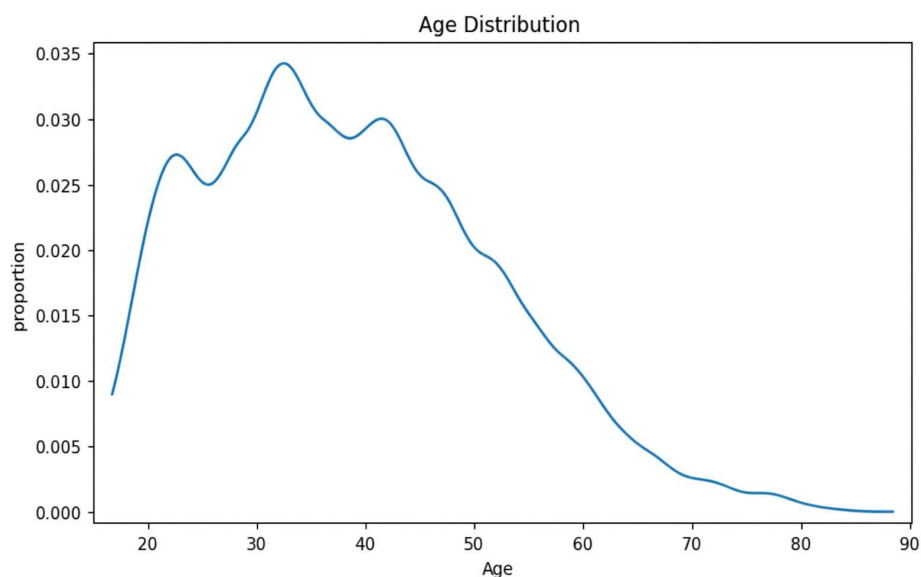
Smooth Tradeoff (right): Consistent descent with moving-average smoothing. Fairness drops  $\sim 34\%$  from  $\lambda=0$  to  $\lambda=6$ . Plateau at  $\lambda \in [3, 5]$  — diminishing returns.

**Conclusion:  $\lambda=2$  is the sweet spot — captures most fairness improvement with minimal utility sacrifice.**

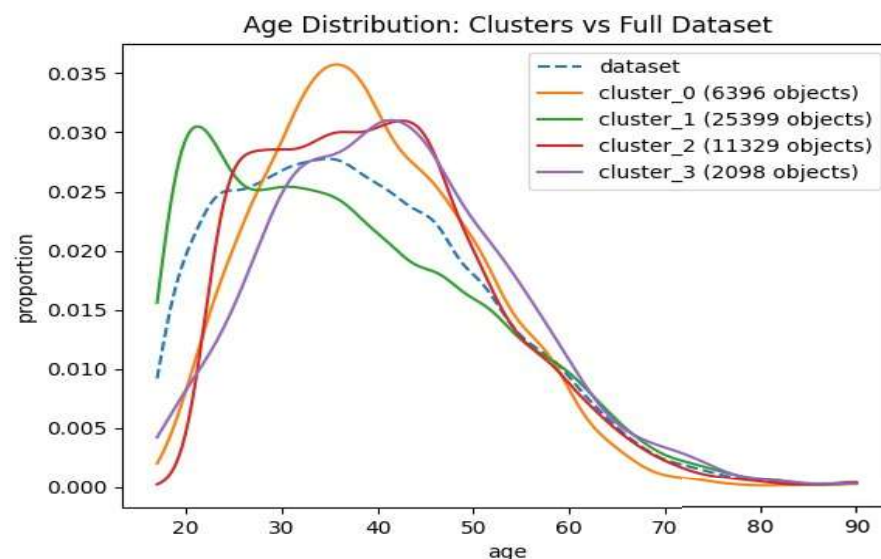
## Results — Age Distribution Analysis

### Age Distribution: Before & After Fairness Constraint

Standard K-Means



Fair K-Means ( $\lambda=2.5$ )



- Standard: Cluster 1 (25,399 pts) over-represents young adults 20–35. Cluster 3 (2,098 pts) is very small.
- Fair: Clusters more balanced — Cluster 0 shrinks to 6,396, redistribution reduces peak at 35 for orange cluster.
- All fair-cluster curves converge closer to the dashed global distribution — KL objective working correctly.



### Points Distribution: Before & After Fairness Constraint

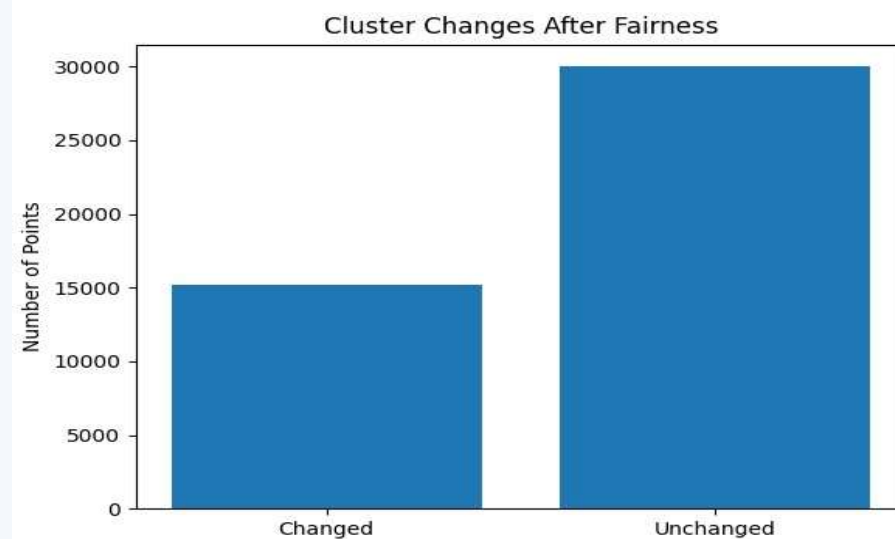
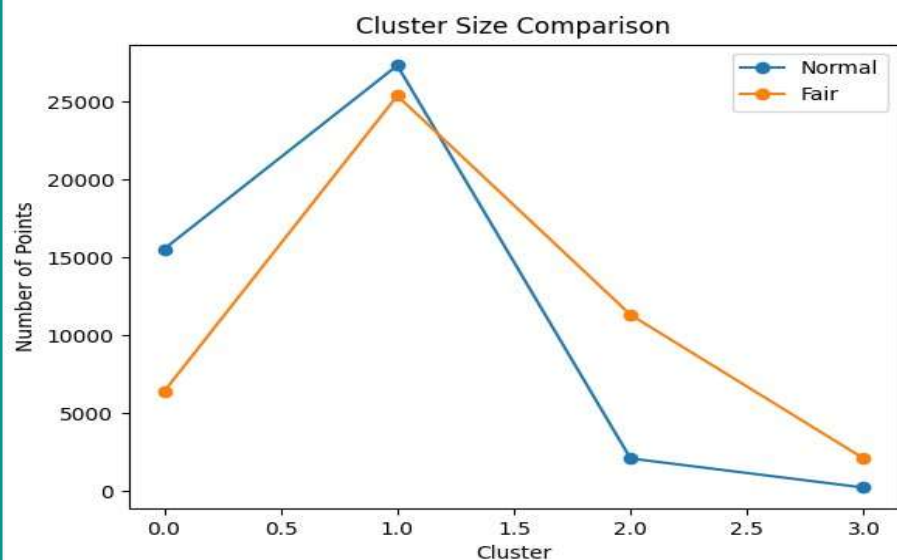
| 1  | index,cluster |
|----|---------------|
| 2  | 0,2           |
| 3  | 1,2           |
| 4  | 2,1           |
| 5  | 3,1           |
| 6  | 4,2           |
| 7  | 5,2           |
| 8  | 6,1           |
| 9  | 7,1           |
| 10 | 8,2           |
| 11 | 9,2           |
| 12 | 10,0          |
| 13 | 11,2          |
| 14 | 12,2          |
| 15 | 13,2          |
| 16 | 14,1          |
| 17 | 15,1          |
| 18 | 16,1          |
| 19 | 17,1          |
| 20 | 18,2          |

- Each point is assigned to the cluster with minimum distance + fairness cost.
- Fairness constraint causes reassignment of some points.
- Final clusters represent balanced and meaningful groupings.



## Results — Cluster Reorganisation

### How Fairness Constraint Reshapes Clusters



- Cluster Size (left): Cluster 0 grows from 15,600 → 6,400; Cluster 2 grows from 2,100 → 11,400. Fairness redistributes.
- Point Changes (right): 15,232 points (33.5%) changed cluster assignments when fairness was applied — significant reorganisation.
- ~30,000 points (66.5%) remained in their original cluster — core cluster structure preserved; only outlier age-groups reassigned.



# Limitations & Future Directions



### Scalability

Current per-point assignment is  $O(nk)$  per iteration. Future work: mini-batch or approximate nearest-neighbour assignment for large-scale datasets.



### Multiple Sensitive Attributes

Extend to joint fairness over multiple attributes (age, gender, income) simultaneously using a multivariate KL or mutual information penalty.



### Adaptive $\lambda$ Selection

Auto-tune  $\lambda$  per dataset using cross-validation or Pareto-optimal front identification, removing need for exhaustive sweep.



### Beyond k-Means

Apply the KL fairness penalty to Gaussian Mixture Models, spectral clustering, or hierarchical methods for richer fairness-aware clustering.



## References

### Key References

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# Thank You

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*Variational Fair Clustering with KL Divergence*

Questions & Discussion

